VERA-DEMO-01 Datasheet

Revision A. This is a fictional board and all specifications are demo data for VERA.

Supply and reset

The board operates from a regulated 3.3 V supply (3.15 V to 3.45 V). Typical idle current is 18 mA. Maximum demo load current is 120 mA. Keep RESET low for at least 10 ms after the supply stabilizes.

Pin assignment

PB6 is I2C1 SCL and PB7 is I2C1 SDA. Both require external 4.7 kohm pull-ups to 3.3 V. PA5 drives the status LED through a 330 ohm resistor. UART debug uses PA9 TX and PA10 RX at 115200 baud, 8 data bits, no parity, 1 stop bit.

I2C electrical timing

Standard mode runs at 100 kHz and fast mode at 400 kHz. The demo bus capacitance must stay below 100 pF. The external sensor uses seven-bit address 0x48. Its ID register at 0x00 returns 0xA1 on revision A.

Clock and temperature

The internal oscillator starts at 16 MHz. The default peripheral clock is 16 MHz. The fictional operating temperature range is 0 to 70 degrees C. Clock accuracy and temperature limits are illustrative, not hardware guarantees.